Project Design Phase
Problem – Solution Fit Template

Date	18 June 2026
Team ID	
Project Name	ShopEZ
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

Template:

References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe

Problem – Solution Fit

1. **Customer Segment(s)** Independent small-to-medium retail vendors and boutique shop owners who want an accessible way to digitize their business without complex technical barriers, alongside modern, tech-conscious mobile shoppers who prioritize ultra-fast, responsive browsing and transparent checkout flows on the move.

2. **Jobs-to-be-Done / Problems** Store operators need to seamlessly transition their physical inventories online, configure front-page layout details and promotional banners without writing code, and track active business counts from a centralized area. Mobile consumers need to quickly search, filter, and buy goods via Cash-on-Delivery (COD) pathways without losing their active shopping cart session data or dealing with hidden checkout fees.

3. **Triggers** Retail merchants are driven to seek a better software alternative when local brick-and-mortar storefront foot traffic drops unexpectedly, when consumer order queries over scattered WhatsApp messaging histories become chaotic and unmanageable, or when institutional network firewalls cause live database connection lookups to time out.

4. **Emotions: Before / After**

Before: Digitally excluded, completely intimidated by complex database configurations or technical jargon, and highly anxious about losing loyal neighborhood customers to massive online retail giants; mobile shoppers feel impatient and untrusting due to heavy application loading delays or unclear transaction values.

After: Confident, independent, and fully in control of what web visitors see on their storefront; small business owners feel relieved by integrated visual metrics, while consumers are highly satisfied with an instantaneous, clear single-page checkout experience.

5. **Available Solutions** Current alternatives are heavy, high-cost enterprise systems (such as Shopify or full-scale Magento setups) that carry steep learning curves, intensive database setups, and expensive developer overhead, contrasted against completely unorganized manual workflows like paper ledger journals, chaotic direct message logs, and basic static catalog pages.

6. **Customer Constraints** Extreme time limitations dictate that boutique operators carry thin cognitive budgets and can only dedicate 1 to 2 hours per day to storefront management amidst sourcing and fulfillment. Hardware limitations mean the system must execute smoothly on budget laptops and mobile screens over cellular data without downloading massive, bloated utility styling frameworks. Financial constraints restrict them to cost-effective, free-tier hosting footprints.

7. **Behaviour** Retail merchants default to tracking stock manually, capturing item pictures to post across social media timelines by hand, and managing fulfillments through direct messages, which frequently leads to lost client details. Shoppers show an incredibly low tolerance for laggy frameworks or multi-tab layout clutter,

immediately abandoning their transactions if an interface fails to remain fast and responsive.

8. **Channels of Behaviour**

Online: Small business forums (such as Reddit r/smallbusiness and r/ecommerce), Instagram micro-merchant communities, and digital threads discussing open-source platform alternatives.

Offline: Local neighborhood craft fairs, regional physical vendor trade markets, small retail merchant associations, and community entrepreneurial meetups.

9. **Problem Root Cause** Most legacy e-commerce structures prioritize heavy, feature-dense corporate frameworks that result in overloaded user control panels, client-side script compilation bloat, and rigid database templates. Furthermore, they completely lack flexible in-memory mock proxies or automated file-based local database fallback routines to keep storefront applications up and running when cloud network access drops.

10. **Your Solution** ShopEZ is a lightweight, high-performance single-page MERN stack application built using highly optimized, framework-free global Vanilla CSS variables to ensure instant mobile rendering. It features a dynamic theme injector that automatically appends an .admin-theme class directly to the document body to instantly remap the interface into a high-contrast dark dashboard layout, enforces transparent itemized pricing receipts, and delivers absolute runtime resilience via a custom database proxy pattern (makeModelProxy) that seamlessly cascades operations onto local file-based JSON storage if cloud connections fail.